

PROJECT EXPO-2026

SMART EV HEALTH MONITORING SYSTEM

Team members:

ECE PROJECT TEAM :

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Voltage • Current • Temperature
Battery Health

Arduino-based monitoring

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Why battery monitoring is needed

- EV batteries need continuous monitoring for safe and reliable operation.
- Changes in voltage, current and temperature can indicate battery stress or abnormal conditions.
- Manual checking is difficult during vehicle operation.
- A low-cost student-friendly system can show battery condition in real time.
- Early warning can help the user take action before unsafe conditions develop.

SYSTEM GOAL

Monitor key battery parameters and provide a simple health indication.

TARGET OUTPUT

**Live values on OLED
+ warning buzzer
+ battery health status**

Existing approaches

- Battery monitoring systems commonly measure voltage, current and temperature.
- Protection systems use measured values to detect abnormal battery conditions.
- IoT-based systems can send battery information to a mobile or remote display.
- Commercial EV battery systems can be costly and complex for student prototypes.

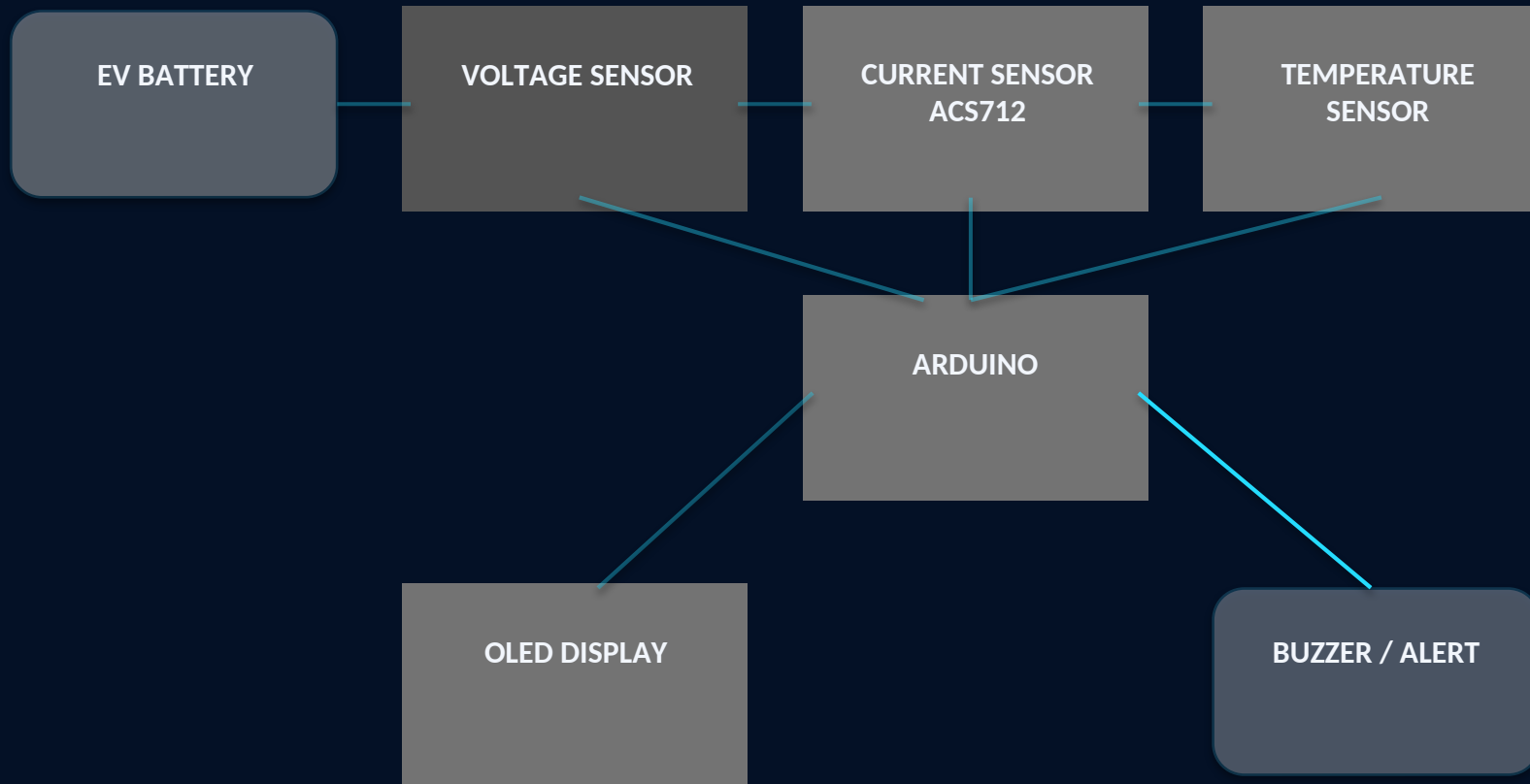
Proposed approach

- Use Arduino as the main controller.
- Measure battery voltage and current with suitable sensors.
- Measure battery temperature using a waterproof temperature sensor.
- Display readings and health status on an OLED.
- Use a buzzer for abnormal-condition warning.

Research gap: a simple, low-cost monitoring prototype combining key battery parameters in one display.

BLOCK DIAGRAM & WORKING

04



Working: Sensors measure battery voltage, current and temperature → Arduino processes the readings → OLED shows the values and battery health → buzzer warns when a limit is exceeded.

Hardware

- Arduino – main controller for reading sensors and processing data.
- Voltage sensor – measures the battery voltage.
- ACS712 current sensor – measures battery current.
- Waterproof temperature sensor – monitors battery temperature.
- 0.96-inch OLED – displays live readings and health status.
- Buzzer – gives an alert during abnormal conditions.

Software & monitoring

- ArduinoIDE
- Embedded Arduino program
- Real-time sensor reading
- Battery health calculation using threshold values
- OLED-based user interface

Example display: V • A • °C • HEALTH

Expected prototype output

- Battery voltage is read continuously.
- Charging/discharging current is monitored.
- Battery temperature is displayed in real time.
- A health percentage/status can be estimated from voltage and operating limits.
- OLED provides a simple local view of battery condition.

OLED MONITORING VIEW

VOLTAGE	12.1 V
CURRENT	1.8 A
TEMP	31 °C
HEALTH	GOOD

Live values shown on OLED
with warning when limits are exceeded.

Functional testing checklist

Voltage sensing	Battery voltage reading appears on OLED	PASS / UPDATE AFTER TEST
Current sensing	ACS712 output changes with load current	PASS / UPDATE AFTER TEST
Temperature sensing	Temperature reading changes with sensor condition	PASS / UPDATE AFTER TEST
OLED display	Voltage, current, temperature and health are displayed	PASS / UPDATE AFTER TEST
Buzzer alert	Buzzer activates when a set limit is crossed	PASS / UPDATE AFTER TEST
Overall system	Arduino reads sensors and updates display continuously	PASS / UPDATE AFTER TEST

Note: Replace the status text with your actual Phase-5 test results after hardware validation.

Limitations

- Prototypemeasurements depend on sensor accuracy and calibration.
- Threshold-based health estimation is a simplified indicator, not a complete battery-management system.
- Actual EV batteries require protection designed for their specific chemistry and voltage range.
- Extensive real-vehicle testing is required before practical deployment.

Future scope

- Add IoT/mobilemonitoring for remote status viewing.
- Store battery history and generate health trends.
- Improve state-of-charge and state-of-health estimation.
- Add automatic charging/discharging protection.
- Develop a compact enclosure for vehicle installation.

Smart monitoring can support safer and more informed EV battery operation.